Class XII Session 2025-26
Subject - Mathematics
Sample Question Paper - 5

Time Allowed: 30 Minutes

General Instructions:

1. This Question paper contains - five sections A,
2. Section A has 18 MCQ's and 02 Assertion-Reason based questions of 1 mark each.

Section A

1. If A = $\begin{bmatrix} \cos a & -\sin a \\ \sin a & \cos a \end{bmatrix}$, and A + A' = I, if the value of a is **[1]**

   a) π
   b) $\frac{3\pi}{2}$
   c) $\frac{\pi}{6}$
   d) $\frac{\pi}{3}$

2. For the set of linear equations **[1]**

   λx - 3y+ z = 0

   x + λy + 3z = 1

   3x + y + 5z = 2

   the value of λ, for which the equations does not have unique solution is

   a) 1, $\frac{11}{5}$
   b) -1, $\frac{-11}{5}$
   c) -1, $\frac{11}{5}$
   d) $-\frac{11}{5}$, 1

3. Family y = Ax + A^3 of curves is represented by the differential equation of degree: **[1]**

   a) 2
   b) 1
   c) 4
   d) 3

4. Differentiation of the following w.r.t. xy = e^{x^3} **[1]**

   a) $x^2e^{x^2}$
   b) $x^3e^{x^3}$
   c) $3x^2e^{x^3}$
   d) $x^2e^{x^3}$

5. The direction ratios of the line x - y + z - 5 = 0 = x - 3y - 6 are proportional to **[1]**

   a) 2, - 4, 1
   b) $\frac{3}{\sqrt{14}}, \frac{1}{\sqrt{14}}, \frac{-2}{\sqrt{14}}$

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c) 3, 1, - 2

d) $\frac{2}{\sqrt{41}}, \frac{-4}{\sqrt{41}}, \frac{1}{\sqrt{41}}$

6. The order and degree (if defined) of the differential equation, $\left(\frac{d^2y}{dx^2}\right)^2 + \left(\frac{dy}{dx}\right)^3 = x\sin\left(\frac{dy}{dx}\right)$ respectively are: **[1]**

a) 2, degree not defined

b) 2, 2

c) 2, 3

d) 1, 3

7. Solution of LPP maximize Z = 2x - y subject to x + y ≤ 2, x, y ≥ 0 **[1]**

a) 4

b) 2

c) 0

d) 1

8. What is the domain of the function $\cos^{-1}(2x - 3)$? **[1]**

a) (-1, 1)

b) [1, 2]

c) [-1, 1]

d) [1, 3]

9. $\int x\sqrt{x^2 - 1}dx = ?$ **[1]**

a) $\frac{1}{3}(x^2 - 1)^{3/2} + C$

b) $\frac{1}{2\sqrt{x^2-1}} + C$

c) $\frac{2}{3}(x^2 - 1)^{3/2} + C$

d) $\frac{1}{\sqrt{x^2-1}} + C$

10. The matrix $\begin{bmatrix} 0 & 5 & -7 \\ -5 & 0 & 11 \\ 7 & -11 & 0 \end{bmatrix}$ is **[1]**

a) an upper triangular matrix

b) a skew-symmetric matrix

c) a symmetric matrix

d) a diagonal matrix

11. The corner points of the feasible region determined by the following system of linear inequalities: **[1]**
2x + y ≤ 10, x + 3y ≤ 15, x, y ≥ 0 are (0, 0), (5, 0), (3, 4) and (0, 5). Let Z = px + qy, where p, q ≥ 0. Condition on p and q so that the maximum of Z occurs at both (3, 4) and (0, 5) is

a) p = q

b) q = 3p

c) p = 3q

d) p = 2q

12. Let $f(x) = x^2|x|$ then the set of values, where f(x) is three times differentiable, is **[1]**

a) 2

b) Infinite

c) Definite

d) 3

13. Find the angle between the following pairs of lines: $\vec{r} = 2i - 5j + k + \lambda(3i + 2j + 6k)$ and $\vec{r} = 7i - 6k + \mu(i + 2j + 2k), \lambda, \mu \in R$ **[1]**

a) $\theta = \cos^{-1}(\frac{19}{21})$

b) $\theta = \sin^{-1}\left(\frac{19}{21}\right)$

c) $\theta = \cot^{-1}\left(\frac{19}{21}\right)$

d) $\theta = \tan^{-1}\left(\frac{19}{21}\right)$

14. In a certain town, 40% persons have brown hair, 25% have brown eyes, and 15% have both. If a person selected at random has brown hair, the chance that a person selected at random with brown hair is with brown eyes **[1]**

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a) $\frac{2}{3}$

b) $\frac{1}{3}$

c) $\frac{3}{8}$

d) $\frac{3}{20}$

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15. The general solution of the differential equation $\frac{dy}{dx} = e^{x+y}$ is **[1]**

a) $e^x + e^y = C$

b) $e^x + e^{-y} = C$

c) $e^{-x} + e^y = C$

d) $e^{-x} + e^{-y} = C$

16. The values of a for which the function f(x) = sin x - ax + b increases on R are **[1]**

a) $(-\infty, -1)$

b) $(-\infty, \infty)$

c) [-1, 1]

d) [1, 1]

17. If $x = at^2$,y = 2 a t, then $\frac{d^2y}{dx^2} =$ **[1]**

a) $\frac{1}{t^2}$

b) 0

c) $-\frac{1}{2a\,t^3}$

d) $\frac{1}{2t^2}$

18. The equation of a line passing through point (2, -1, 0) and parallel to the line $\frac{x}{1} = \frac{y-1}{2} = \frac{2-z}{2}$ is: **[1]**

a) $\frac{x-2}{1} = \frac{y-1}{2} = \frac{z}{2}$

b) $\frac{x-2}{1} = \frac{y+1}{2} = \frac{z}{-2}$

c) $\frac{x+2}{1} = \frac{y-1}{2} = \frac{z}{-2}$

d) $\frac{x+2}{1} = \frac{y-1}{2} = \frac{z}{2}$

19. **Assertion (A):** The Points A(2, 3, -4), B(1, -2, 3) and C(3, 8, -11) are conllinear **[1]**

Reason (R): It direction ratios of AB and BC are proportional, A, BC are collinear points.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

20. **Assertion (A):** $\int \frac{dx}{16+9x^2} = \frac{1}{12}\tan^{-1}\left(\frac{3x}{4}\right)$ **[1]**

Reason (R): $\int \frac{1}{a^2+x^2}dx = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + C$

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

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